Kate Windsor

Education

PhD Candidate in Cell and Molecular Biology — **May 2028 (Expected)**
The University of Texas at Austin

B.S. in Neuroscience and Spanish, Minor in Religion — **May 2019**
Trinity University, San Antonio, Texas
GPA: 3.52, *Cum Laude*

Research Publications

- Ghatpande, V., Lewis, A. J., **Windsor, K. E.**, Lee, H., Alcala, A. J., Menke, D. B., Cenik, C. & Vokes, S. A. (2026). Developmental and transcriptional programs that define the initiation of the forelimb. *Development 153*(14).
- Ramachandran, J., Chen, W., Lex, R. K., **Windsor, K. E.**, Lee, H., Wang, T., Zhou, W., Ji, H., & Vokes, S. A. (2023). The BAF chromatin complex component SMARCC1 does not mediate GLI transcriptional repression of Hedgehog target genes in limb buds. *Developmental Biology*, 504, 128–136.
- Lex, R. K., Zhou, W., Ji, Z., Falkenstein, K. N., Schuler, K. E., **Windsor, K. E.**, Kim, J.D., Ji, H., & Vokes, S. A. (2022). GLI transcriptional repression is inert prior to Hedgehog pathway activation. *Nature communications*, *13*(1), 1-15.
- Moreno Manrique, J. F., Voit, P. R., **Windsor, K. E.**, Karla, A. R., Rodriguez, S. R., & Beaudoin III, G. M. (2021). SynapseJ: an automated, synapse identification macro for ImageJ. *Frontiers in Neural Circuits*, 97.

Research Presentations

- **Windsor, K.** (2026, October 15). *Manganese-induced parkinsonism and career talk.* [Skype-a-Scientist presentation; High School]. Sidwell Friends School, Washington DC, United States.
- **Windsor, K.** (2026, October 5, 6, 7. 2025, February 21; 2024, November 7). *Manganese and Your Brain*. [Present your PhD to a 12-year-old presentation; Middle School]. Kealing Middle School, Austin, Texas, United States.
- **Windsor, K.** (2026, September 24). *The Role of HIF Signaling in Manganese-induced Parkinsonism*. [Department presentation]. 4th Year Graduate Student Research Showcase, Austin, TX, United States.
- **Windsor K.**, Jursa T, Smith DR, Mukhopadhyay S. (2026, July 23-26). *Dopaminergic HIF2α is required for manganese-induced neuromotor dysfunction.* [Poster Presentation]. Annual meeting of the Society for Developmental Biology, Las Vegas, Nevada, United States.
- **Windsor K.**, Jursa T, Smith DR, Mukhopadhyay S. (2026, June 8-11). *Dopaminergic HIF2α is required for manganese-induced neuromotor dysfunction*. [Poster Presentation and Flash Talk]. Trace Elements in Biology and Medicine, Nashville, Tennessee, United States.
- **Windsor, K.**, Jursa, T., Smith, D. R., Mukhopadhyay, S. (2026, March 22-26). *Role of brain HIF signaling in motor function and manganese neurotoxicity.* [Poster Presentation]. Society of Toxicology, San Diego, California, United States.
- **Windsor, K.** (2026, January 13). *Biomagnification*. [Skype-a-Scientist presentation; 4th grade]. Quaker Ridge School, Scarsdale, New York, United States.
- **Windsor, K.** (2025, May 5). *How Your Brain Grows and Learns*. [Elementary school presentation]. Lee Elementary, Austin, Texas, United States.

- **Windsor, K.** (2025, April 25). *Manganese and Your Brain*. [Present your PhD to a 12-year-old presentation; All Ages]. Austin Homeschoolers Science Club, Austin, Texas, United States.
- **Windsor, K.** (2024, September 13-15). *Dissecting Mechanisms of Neuronal Manganese Homeostasis*. [Department presentation]. Interdisciplinary Life Science Graduate Program Annual Retreat, Austin, TX, United States.
- **Windsor, K.**, Voit, P., Litch, A., Hemani, S., Karla, A., Guo, C., Ferrer, P., & Beaudoin III., G. M. (2018, November 3-7). *Structural and functional changes of input-specific excitatory synapses on nigral dopamine cells caused by cocaine.* [Poster Presentation]. Society for Neuroscience, San Diego, California, United States.

Other Publications & Media

- **Windsor, K.** (2026, October 22). Under antibiotic siege, frontline bacterial cells protect their colony from beyond the grave. *College of Natural Sciences, University of Texas at Austin.* https://cns.utexas.edu/news/research/under-antibiotic-siege-frontline-bacterial-cells-protect-their-colony-beyond-grave
- **Windsor, K.** (2026, October 15). Report from the Society for Developmental Biology 85th Annual Meeting. *Society for Developmental Biology*. https://www.sdbonline.org/page.cfm?name=resource&ResourceID=4627
- **Windsor, K.** (2026, August 8). Alexandra Joyner Awarded 2026 Society for Developmental Biology Lifetime Achievement Award. *Society for Developmental Biology*. https://www.sdbonline.org/resource?ResourceID=4597
- **Windsor, K.** (2026, April 23). Debating the Benefits and Risks of Plastic. *Society of Toxicology*. https://toxchange.toxicology.org/blogs/kate-windsor/2026/04/23/debating-the-benefits-and-risks-of-plastic
- **Windsor, K.** (2026, April 18). Hungarian Mummies of the White Plague: Homeotic Transformation in a Historic Epidemic. *Developmental Biology*, *536*, 204-205. https://doi.org/10.1016/j.ydbio.2026.04.009
- **Windsor, K.** (2025, December 23). How do electric eels work? New observations on the world's oldest (organic) battery. *Developmental Biology, 532,* 158-159. https://doi.org/10.1016/j.ydbio.2025.12.019
- **Windsor, K.** (2025, November 21). A Dash for the finish line, with help from GetHIRED! *Society for Developmental Biology*. https://www.sdbonline.org/resource?ResourceID=4534
- **Windsor, K.** (2025, August 11). UT science communication is by students, for students. *The Daily Texan*. https://thedailytexan.com/2025/08/11/ut-science-communication-is-by-students-for-students/
- **Windsor, K.** (2025, August 8). Autism research should not be at the mercy of RFK Jr.'s ignorance. *The Daily Texan.* https://thedailytexan.com/2025/08/08/autism-research-should-not-be-at-the-mercy-of-rfk-jr-s-ignorance/
- **Windsor, K.** (2025, July 21). Revolution is driven by art. *The Daily Texan.* https://thedailytexan.com/2025/07/21/revolution-is-driven-by-art/
- **Windsor, K.** (2025, July 13). Cognitive offloading: why do more, when you could do less? *The Daily Texan.* https://thedailytexan.com/2025/07/13/cognitive-offloading-why-do-more-when-you-could-do-less/
- **Windsor, K.** (2025, July 7). Academia isn't dead, but it's losing ground. *The Daily Texan.* https://thedailytexan.com/2025/07/07/academia-isnt-dead-but-its-losing-ground/
- **Windsor, K.** (Host). (2024). *She Blinded Me with Science!* [Audio podcast]. KVRX. https://shows.acast.com/sciencekvrx

Research Experience

Graduate Research Assistant **January 2024 – Present**

University of Texas, Austin, TX
Supervisor: Dr. Somshuvra Mukhopadhyay

- Investigate the role of hypoxia inducible factor isoforms in manganese-induced parkinsonian syndrome
- Manage multiple mouse lines, including daily treatments and behavioral testing

Research Assistant **August 2020 – June 2023**
University of Texas, Austin, TX
Supervisor: Dr. Steven Vokes

- Conducted primary research in embryonic limb buds to elucidate mechanisms of the *Hedgehog* signaling pathway
- Maintained mouse colony and breed new transgenic lines
- Managed lab inventory and ordering to support ongoing experiments

Research Assistant **June 2019 – July 2020**
Austin Speech Labs, Austin, TX
Supervisor: Shilpa Shamapant

- Designed and implemented experimental therapy conditions to increase speech recovery in stroke survivors with Broca's aphasia
- Administered bilingual speech therapy to measure effect of therapy language on anomia rehabilitation
- Recorded speech ability assessment scores and created visual representations of clients' progress

Undergraduate Research Assistant **August 2017 – May 2019**
Trinity University, San Antonio, TX
Department of Biology
Advisor: Dr. Gerard Beaudoin III

- Conducted primary research on structural changes of input-specific excitatory synapses on nigral dopamine cells caused by cocaine
- Developed ImageJ macro to analyze colocalization and intensity of signal
- Mentored undergraduate students in lab techniques

Undergraduate Research Assistant **August 2016 – May 2017**
Trinity University, San Antonio, TX
Department of Biology
Advisor: Dr. James Roberts

- Maintained cell lines to investigate role of estrogens in mediating neural protection and recovery from damage due to oxidative stress

Undergraduate Research Assistant **August 2016 – December 2016**
Trinity University, San Antonio, TX
Department of Psychology
Advisor: Dr. William Ellison

- Recruited and coordinated undergraduate research participants
- Encoded participant data to model heterogeneity and stability of dynamic factors in within-person variability of personality

Awards

- **Best Graduate Student Presentation** *(Out of 300)*, Society for Developmental Biology **July 2026**
- **Dr. Marsha Wheeler Miller Fellowship Recipient** **May 2024**

- **Alamo Brain Bowl Grand Champion**, Trinity University Brain Bowl Team **Spring 2019**
- **Trainee Professional Development Award**, Society for Neuroscience **September 2018**

Service & Outreach

- **Volunteer Scientist** **December 2025 - Present**
  Skype-a-Scientist (International STEM Outreach Program)
- **President** **May 2025 - Present**
  Present your PhD to a 12-year-old, University of Texas at Austin
- **Judge** **September 2025**
  Fall Undergraduate Research Symposium Science Slam, University of Texas at Austin
- **Volunteer Pen Pal** **September 2024 - Present**
  Letters to a Pre-Scientist (National STEM Outreach Program)
- **Science Fair Judge**, 4th – 7th Grade **March 2019**
  St. George Episcopal School, San Antonio, TX
- **President, Treasurer** **April 2016 – December 2018**
  Neuroscience Student Association, Trinity University

Teaching Experience

- **Co-Teacher**, Infectious Disease and the Arts **Summer 2026**
  Texas Prison Education Initiative, Coleman Unit, Lockhart, TX
- **Teaching Assistant**, Integrative Biology Lab **Spring 2019**
  Department of Biology, Trinity University, San Antonio, TX
- **Teaching Assistant**, 1st – 8th Grade Science **Summer 2017**
  Department of Science, Real Colegio Nuestra Señora del Loreto, Madrid, Spain

Skills

- Molecular Biology Techniques: Immunofluorescence, Light & Confocal microscopy, Cryostat sectioning, PCR genotyping, RNA isolation, qPCR, Skeletal preparation, Western blot, Cell culture, Molecular cloning
- Animal techniques: Mouse handling, Breeding, Colony management, Behavioral assays, Euthanasia, Dissection, Intraperitoneal and subcutaneous injection, Perfusion
- Data analysis: Microsoft Excel, SPSS, ImageJ, R, Prism
- Fluent in Spanish